if the receiver gets closer to A than $l_{\min}$, the defender at B cannot intercept (Fig. 2). If $L \leqslant l_{\min}$, it is part 2. If $L > l_{\min}$, let the receiver be at E on Ax.

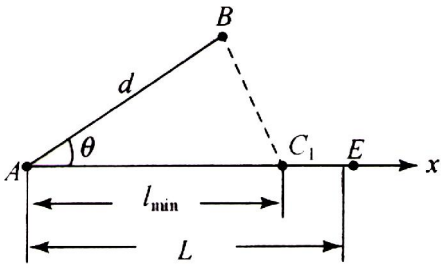


Fig. 2

His time to reach C_1 is

$$t_{rm} = (L - l_{\min}) / v_r \ ; \qquad (8)$$

Defender arrival time at C_1: $t_{pm} = (d^2 + l_{\min}^2 - 2dl_{\min}\cos\theta)^{1/2} / v_p$.

The condition for avoiding interception is

$$t_{rm} < t_{pm} \ . \qquad (9)$$

Thus $$L < \frac{v_r}{v_p}(d^2 + l_{\min}^2 - 2dl_{\min}\cos\theta)^{1/2} + l_{\min} , \qquad (10)$$

where $l_{\min}$ is given by (5).

Method 2: the defender and ball reach C_1 at the same time, so

$$t_{pm} = l_{\min} / v \ .$$

The no-interception condition (9) and equation (8) give

$$L < (1 + v_r / v) l_{\min} \qquad (11)$$

Again $l_{\min}$ is given by (5). Equations (11) and (10) clearly agree.

2.

1. (I) In a chosen frame, satellite motion gives its coordinates x, y, z at every instant. A signal emitted at t reaches ground station i at t_i. The signal travels at light speed c, hence

$$(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = c^2 (t - t_i)^2 \qquad (i = 1, 2, 3), \qquad (1)$$

x_i, y_i, z_i are known, premeasured station i coordinates; its clock measures t_i. The signal supplies t, also known. Equation (1) thus has three unknowns, x, y, z, requiring three independent equations: at least three ground stations. These are (1) for $i = 1, 2, 3$.

(II) (i) Earth centre O and stations D_1, D_2 form an isosceles triangle; the equal sides are R. Thus

